

AI for Bharat Hackathon

Powered by **aws**



Team Name : **LAZARUS**

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Problem Statement : **AI for Learning & Developer Productivity**

Brief about the Idea:

The Problem

The #1 way developers learn is by reading real codebases — not tutorials. But millions of valuable GitHub projects are trapped in **dead, outdated code** that won't even install anymore. A student finds a perfect full-stack project, runs npm install, hits 47 errors, and gives up. **The knowledge inside that codebase dies with it.**

And developers themselves are victims too — that portfolio project from 2 years ago? That hackathon winner? They're all rotting. Dependencies outdated, **frameworks 3 versions behind, security vulnerabilities everywhere**. Your own past work becomes **legacy code** that embarrasses you instead of showcasing your skills. You don't have weeks to rewrite everything. So it just sits there. **Dead.**

200M+ GitHub repos exist. **Most can't run. Nobody learns from dead code.**

What is Lazarus?

Lazarus is an **AI-powered platform that resurrects any dead or legacy codebase — turning it into a modern, runnable, explorable application that developers can learn from, experiment with, and keep alive.**

Paste any GitHub link — someone else's or your own old repo. Get back:

- **Modernized, working code deployed with a live URL**
- **Side-by-side diffs** — old vs new patterns, so you learn why things changed
- **In-browser AI code editor** — ask questions about any file, get contextual explanations
- **Visual AI Design mode** — click any element in the live preview, describe changes in English, watch code update in real-time
- **One-click PR back to your repo** — keep your own projects permanently up to date

It's an AI tutor + code resurrection engine + developer productivity tool in one.

How different is it from any of the other existing ideas?

Existing Tools	What they do	LAZARUS
Copilot / ChatGPT	File-level code suggestions	Full-repo resurrection + deployment
Codemod/OpenRewrite	Rule-based migrations only	AI understands intent + modernizes safely
Most AI dev tools	Generate code only	Generates + runs + fixes automatically
Traditional modernization	Manual, months, expensive	5-minute autonomous resurrection
Any current system	Static AI	Self-learning AI that improves each run

How will Lazarus solve the problem?

Lazarus eliminates the pain of dead and outdated codebases by *automatically scanning, analyzing, and modernizing any GitHub repository through a 7-phase AI pipeline — fully transparent and interactive at every step*. It detects the tech stack, maps all dependencies and APIs, generates a file-by-file migration plan with reasoning, rewrites every file to modern code in parallel, and deploys the resurrected app with a live working URL. If anything breaks, the self-healing engine reads error logs, diagnoses the issue, patches the code, and re-deploys automatically. Once alive, developers can explore side-by-side diffs to learn why each pattern evolved, ask the AI code editor questions about any file, visually redesign the live app by clicking elements and describing changes in plain English, and send a one-click pull request back to their own repo to keep it permanently up to date — all in one smart, effortless platform for learning and developer productivity.

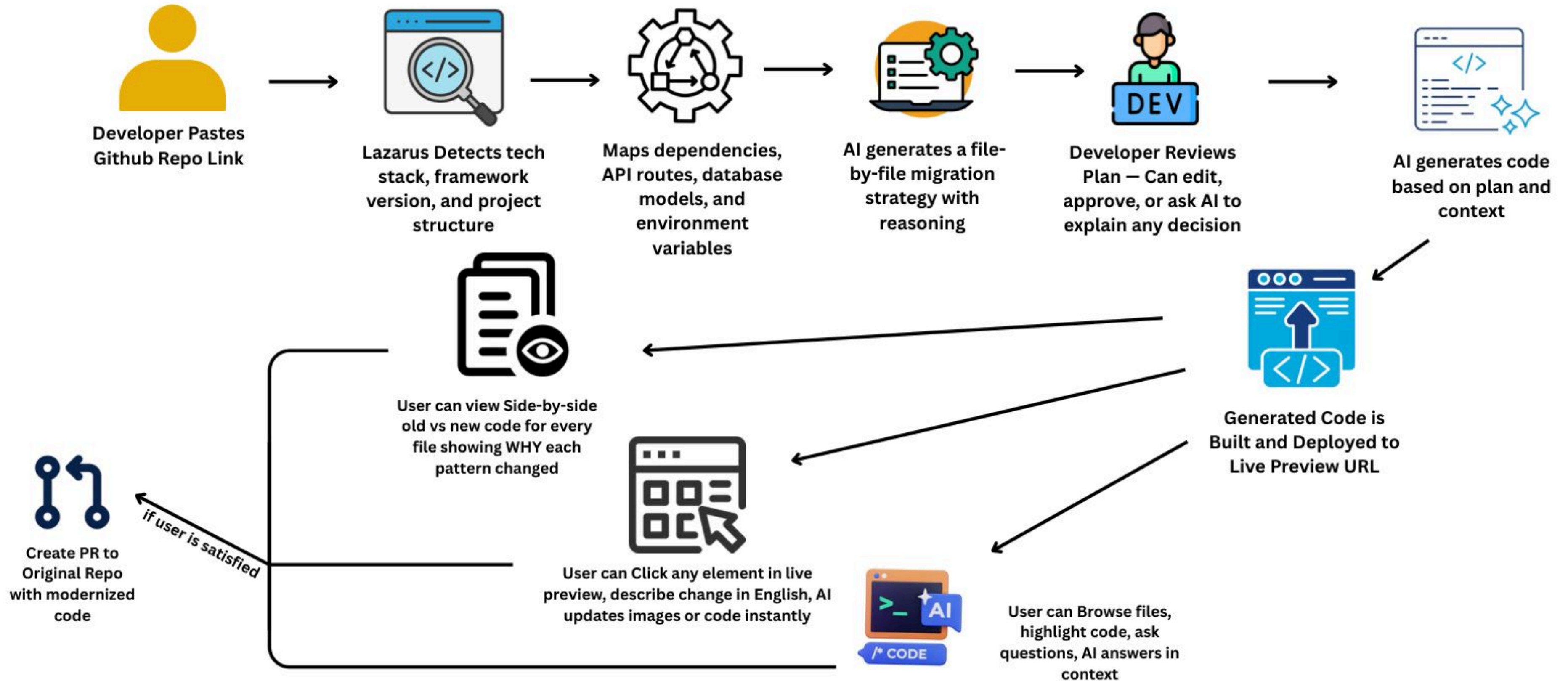
USP of The Proposed Solution

Lazarus is the only platform that delivers *true end-to-end resurrection of legacy codebases by combining AI-powered code understanding, automated migration, and live deployment in a single seamless workflow*. Unlike basic code generators or dependency updaters, Lazarus not only modernizes entire repositories with detailed reasoning for every change, but also provides interactive learning tools—side-by-side diffs, contextual AI explanations, and a visual design mode where developers can experiment by describing changes in plain English and seeing instant results. Its self-healing pipeline automatically diagnoses and fixes errors post-deployment, while one-click pull requests keep personal and project repositories perpetually up to date. All of this happens transparently, empowering developers to learn, modernize, and maintain codebases effortlessly—something no other solution offers.

List of features offered by the solution

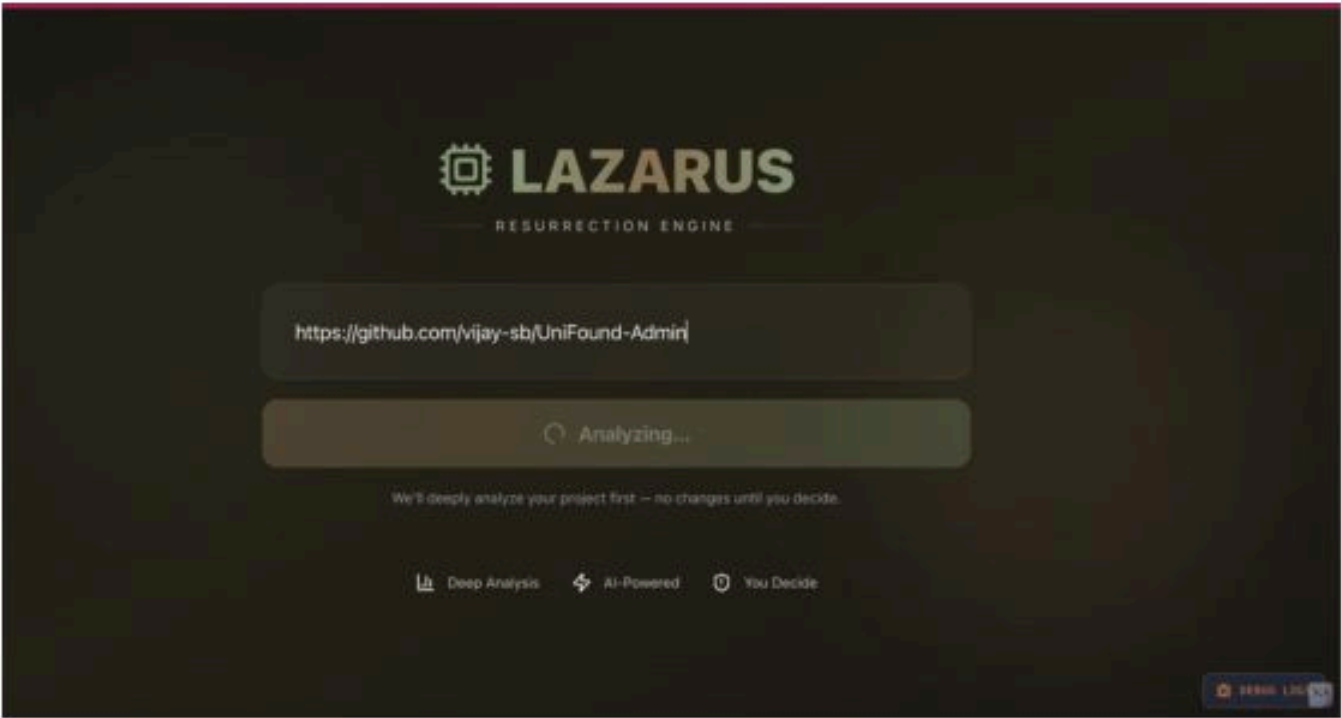
Category	Features
Resurrection Pipeline	One-link resurrection — paste any GitHub URL, zero config needed · Automatic tech stack & dependency detection · AI migration plan with reasoning for every change · Parallel code generation to modern frameworks · Auto build, containerize, and deploy with a live URL · Self-healing — AI reads errors, patches, and re-deploys automatically (up to 5x)
Learning Tools	Side-by-side diff view — old vs new code for every file showing why patterns changed · AI contextual Q&A — highlight any code, ask questions, get answers with full project context · Migration plan doubles as a teaching document explaining every decision
AI Code Editor	In-browser code editor with full file tree navigation · AI-assisted editing with suggestions and explanations · Inline diff accept/reject for every file — developer stays in control
Visual AI Design	Click any element in the live preview to select it · Describe changes in plain English · AI generates images or rewrites code instantly · Live before/after preview — see visual + code change together
Keep Your Repos Alive	Feed your own old repos to modernize them · One-click pull request back to your original GitHub repo · Re-run anytime — portfolio, side projects, past work never become outdated again
Security & Monitoring	Secure authentication and secrets management · Real-time pipeline status via WebSocket · Full build/deploy logs and request tracing · Serverless architecture — ~\$0.70–\$2.50 per resurrection

Process flow diagram or Use-case diagram

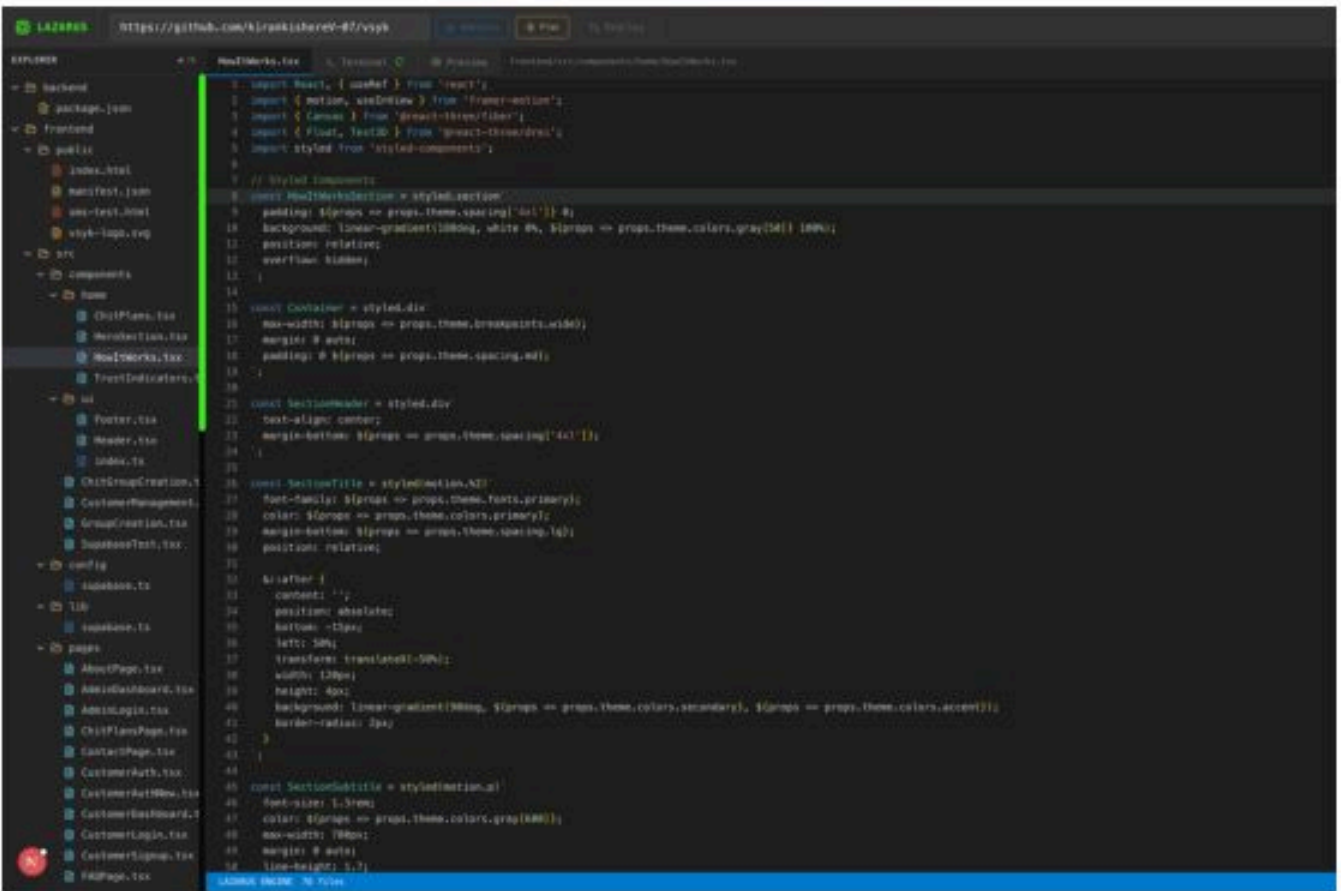


Wireframes/Mock diagrams of the proposed solution (optional)

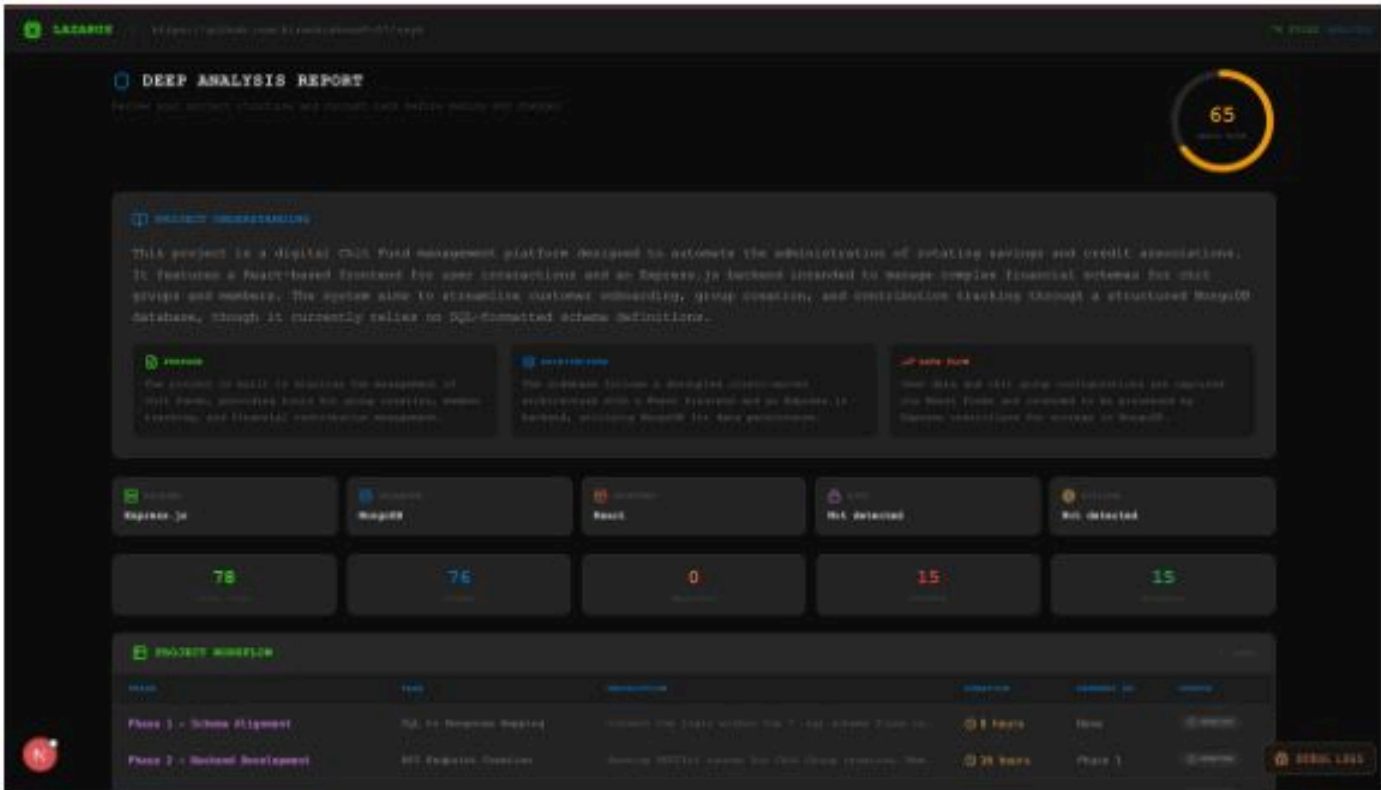
HOME PAGE



CODE VISUALIZER



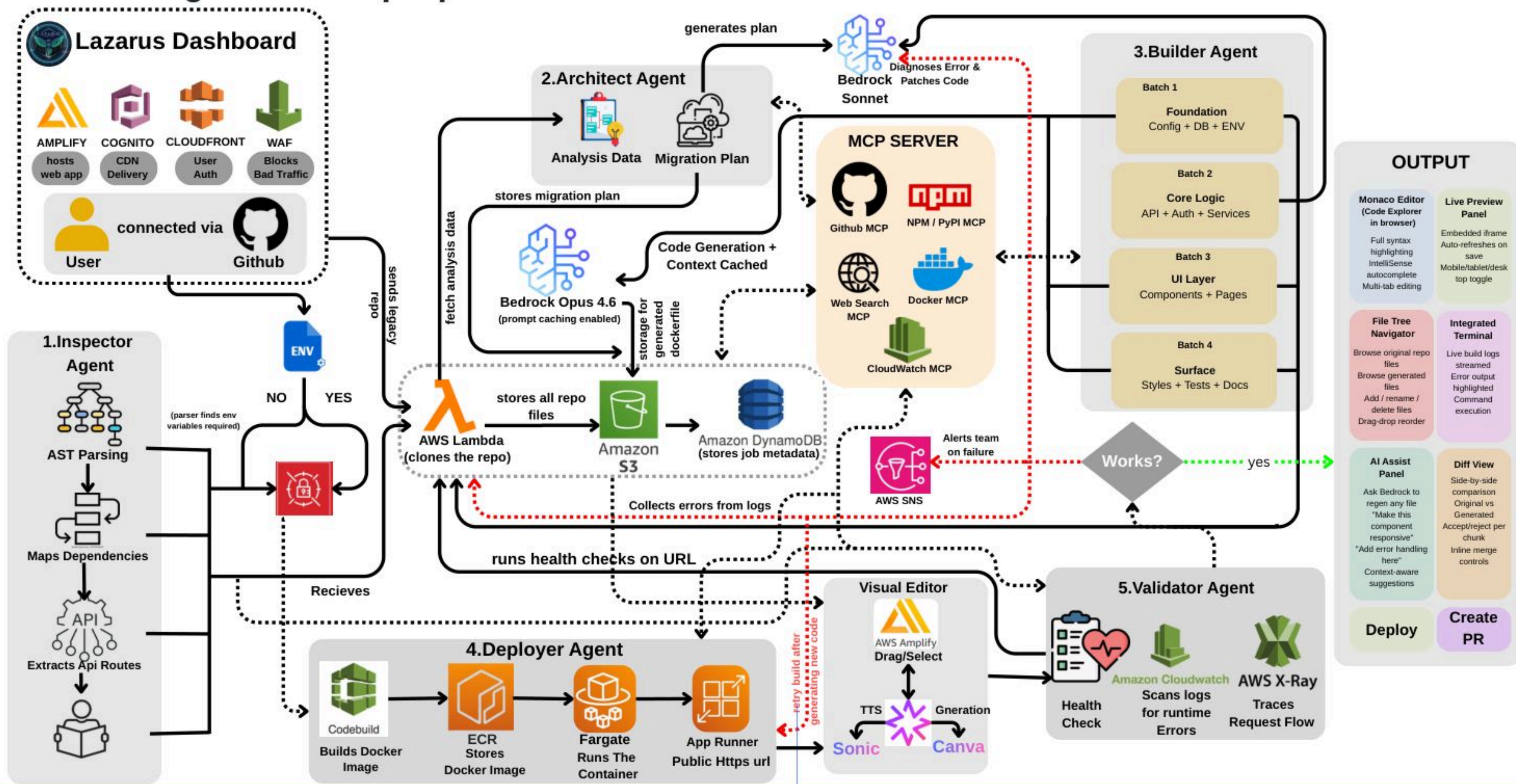
DETAILED ANALYSIS



REQUIRED CHANGES



Architecture diagram of the proposed solution:



Technologies to be used in the solution:

AI TECHNOLOGIES

- **Amazon Bedrock Claude Sonnet**  : Migration planning, code generation, self-healing, code Q&A, Visual AI Design code changes
- **Amazon Bedrock Claude Haiku**  : Tech stack detection, environment variable classification, lightweight analysis
- **Canva Image Gen**  : AI image generation and editing in Visual AI Design mode
- **MCP Servers (Model Context Protocol):**
 - **GitHub MCP**  : Repo access and cloning
 - **NPM/PyPI MCP**  : Registry lookups for dependency analysis
 - **Web Search MCP:** Fetching migration guides and documentation
 - **Docker MCP:** Container builds
 - **CloudWatch MCP:** Log analysis for self-healing

BACKEND

- **AWS Lambda**  : API handlers, pipeline phase execution, validation
- **Amazon API Gateway**  : REST and WebSocket APIs
- **Amazon S3**  : Repo file storage, generated code, assets
- **Amazon DynamoDB**  : Project state, migration plans, user data
- **Amazon ElastiCache**  : Context caching for cost-efficient AI calls
- **AWS CodeBuild**  : Building Docker images for resurrected apps
- **Amazon ECR**  : Docker image registry
- **AWS App Runner**  : Deploying resurrected apps with live URLs
- **ECS Fargate**  : Running MCP servers and long-running tasks
- **AWS Secrets Manager**  : Secure handling of API keys and env variables
- **Amazon CloudWatch**  : Logging and monitoring
- **AWS X-Ray**  : Request tracing
- **Amazon SNS** : Failure notifications and alerts
- **AWS WAF** : Web application firewall for security

FRONTEND

- **Next.js:** React framework for the platform UI
- **Monaco Editor:** In-browser code editor (VS Code engine) for file browsing, editing, and diff view
- **AWS Amplify:** Frontend hosting and deployment
- **Amazon CloudFront:** CDN for fast global delivery
- **Amazon Cognito:** User authentication and session management
- **WebSocket API:** Real-time pipeline status updates via API Gateway

Estimated implementation cost (tentative):

Lazarus is designed to be fully serverless and cost-efficient, using pay-as-you-go AWS services. The estimated cost to fully resurrect and modernize a typical codebase is:

AI Model Usage (Bedrock) : \$0.50 – \$2.00

Lambda, Step Functions, Orchestration: \$0.05 – \$0.10

Build & Deploy (CodeBuild, App Runner, ECR): \$0.10 – \$0.30

Storage (S3, DynamoDB, ElastiCache): \$0.02

Monitoring & Notifications: <\$0.05

TOTAL PER PROJECT: \$0.70 – \$2.50

Summary:

Each full resurrection (scan, migrate, deploy, validate, and self-heal) costs approximately \$0.70 to \$2.50 per project. This is dramatically lower than manual migration, which can cost hundreds to thousands of dollars in developer time. Costs scale only with usage, making Lazarus highly affordable for individuals, teams, and organizations.

